

Momentum Formula Triangle

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS The triangle links momentum (p), mass (m), and velocity (v). Cover the unknown to read the formula: $p = m \times v$, $m = p \div v$, and $v = p \div m$.

Round 1 — Retrieve It

1. What three quantities does the momentum triangle connect?

2. Write the formula for momentum.

3. Write the formula for mass.

4. Which variable do you cover to find velocity, and what formula results?

Round 2 — Sketch It

Draw the momentum triangle. Put momentum on top; put mass and velocity on the bottom. Circle the two quantities you would multiply to find momentum.

Momentum — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. A cart has a mass of 4 kg and moves at 5 m/s. What is its momentum?

2. An object has a momentum of 60 kg·m/s at a velocity of 12 m/s. What is its mass?

3. A 30 kg object has a momentum of 90 kg·m/s. What is its velocity?

Round 4 — Reason from Evidence

A heavy, slow object has a mass of 8 kg and a velocity of 3 m/s. A light, fast object has a mass of 2 kg and a velocity of 10 m/s. Use the triangle to find each object's momentum, then say which has more momentum and how you know.

Momentum — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how the momentum triangle helps you find a missing quantity. Use one example that is not printed on the cue card.

C**Connect**

Connect momentum to why a heavier or faster object is harder to stop. How does that relate to $p = m \times v$?

E**Extend**

Create your own problem that asks for velocity. Give the numbers and show the work using $v = p \div m$.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Momentum

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Momentum, mass, and velocity.
2. $p = m \times v$.
3. $m = p \div v$.
4. Cover v ; $v = p \div m$.

Sketch It — look for

Momentum on top, with mass and velocity on the bottom. Momentum is found by multiplying mass $\times$ velocity (the two bottom quantities).

Apply It

1. $p = m \times v = 4 \times 5 = 20 \text{ kg}\cdot\text{m/s}$.
2. $m = p \div v = 60 \div 12 = 5 \text{ kg}$.
3. $v = p \div m = 90 \div 30 = 3 \text{ m/s}$.

Reason from Evidence — look for

Heavy, slow object: $p = 8 \times 3 = 24 \text{ kg}\cdot\text{m/s}$. Light, fast object: $p = 2 \times 10 = 20 \text{ kg}\cdot\text{m/s}$. The heavy, slow object has more momentum ($24 > 20$) because momentum depends on both mass and velocity.

ACE — Extend look-for

Many answers possible. Check that the problem asks for velocity and the work correctly uses $v = p \div m$.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.